1. **Aim of the practical:** To understand and demonstrate the usage of grep, e-grep, and f-grep commands in Linux for searching and filtering text patterns efficiently within files or streams.

2. **Objective:**

   a) Understand how to use the basic grep command to find patterns in files with regular expressions.
   b) Discover the egrep command, which uses extended regular expressions that include more special characters for matching patterns.
   c) Learn how to use the fgrep command to search for exact strings (without using regular expressions) in files.

   d) Grasp the performance differences and best practices for choosing between these commands based on your needs.

3. **Hardware and Software Requirements:**

   Computer System: Host system running virtualization software.
   Operating System: OS on the virtual machine (e.g., Fedora).

4. Commands/Steps:

Step-1: Touch and cat command to create and access file

.

```
asr@fedora:~$ touch hello.txt
asr@fedora:~$ cat >hello.txt
I am Aaryaman Singh
^C
```

Step-2: Grep commandscat

-- color: By default, this option enables colorized output, highlighting the matching patterns.

```
asr@fedora:~$ grep --color '5th worksheet" hello.txt
> This worksheet is made by Aaryaman Singh
```

Step-3: Grep -r "string" * command: searching for a string recursively in all directories

```
asr@fedora:~$ grep -r "Aaryaman"
story.txt:This is me Aaryaman Singh
grep: .story.txt.swp: binary file matches
```

Step-4: Grep -I "string" command is use to ignore case sensitive

```
asr@fedora:~$ cat >hello.txt
Waning
Error
Failed
^C
asr@fedora:~$ cat > pattern.txt
warning
error
failed
^C
```

Step-5: Grep -c: count the lines where strings are matched with -c option

```
asr@fedora:~$ cat > hello.txt
Aaryaman
Aaryaman
Aaryaman
Aaryaman
^C
asr@fedora:~$ grep -c "Aaryaman" hello.txt
4
```

REJEX COMMANDS:

Step-6: ‘ ^ ’ :matches char at the beginning of the line

```
asr@fedora:~$ grep ^H hello.txt
He is bad boy
```

Step-7: character $ ‘ ’: matches char at the end of the line

```
asr@fedora:~$ cat >hello.txt
tit for tat
^C
asr@fedora:~$ grep 't$' hello.txt
tit for tat
```

Step-8: “.” Matches any character.

```
asr@fedora:~$ grep "." hello.txt
tit for tat
```

Step-9 Step-[a-z] matches any char between a and z.

```
asr@fedora:~$  grep '[a-z]' hello.txt
tit for tat
```

Step-10: ep-[^ ..]Matches anything apart from what is contained in the brackets

```
asr@fedora:~$ cat >hello.txt
hey, 7404729499 tata docomo
^C
asr@fedora:~$ grep '[^a-z'] hello.txt
hey, 7404729499 tata docomo
```

Step-11: F- grep commands

Grep -f (main_file file_to_search): matches and show same pattern from more thanone file.

```
asr@fedora:~$ cat >hello.txt
my linux is good
error
warning
faild
^C
asr@fedora:~$ cat > pattern.txt
error
warning
failed
^C
asr@fedora:~$ grep -f hello.txt pattern.txt
error
warning
```

Step-12: Grep -f (-v) :invert match ….show the lines that do not match the patterns

```
asr@fedora:~$ grep -f pattern.txt -v hello.txt
my linux is good
```

Step-13: Grep -f (-i) : ignore case sensitive over the files

```
asr@fedora:~$ grep -f pattern.txt -i hello.txt
Error
Failed
```

```
asr@fedora:~$ grep -f hello.txt -i pattern.txt
error
failed
```

E-grep commands:

Step-14: Grep -e "pattern1" -e "pattern2" file_name: this commands allows you to specify multiple patterns to search in the input file.

```
asr@fedora:~$ cat >hello.txt
failed to show error in his new directory.
^C
asr@fedora:~$ grep -e "directory." -e "failed" hello.txt
failed to show error in his new directory.
asr@fedora:~$ grep -e "directory." -e "failed" -o hello.txt
failed
directory.
```

Step-15: Grep -e (-o): show only the part of the line that matches

```
asr@fedora:~$ grep -e "directory." -e "failed" -o hello.txt
failed
directory.
```

5. Learning outcomes (What I have learned):

a) Learned to use the grep command for searching specific text patterns within files.
b) Understood the differences between grep, egrep, and fgrep, and when to use each one.
c) Acquired knowledge on applying regular expressions with grep for advanced and flexible text searching.

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature